

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col
```

```
products_data = [
    (1, "Хлеб"),
    (2, "Молоко"),
    (3, "Яблоко"),
    (4, "Сок"),
    (5, "Кофе")
]
categories_data = [
    (1, "Еда"),
    (2, "Напитки"),
    (3, "Фрукты")
]
product_category_data = [
    (1, 1), # Хлеб - Еда
    (2, 1), # Молоко - Еда
    (3, 3), # Яблоко - Фрукты
    (4, 2)  # Сок - Напитки
    # Кофе не имеет категории
]
```

```
class Warehouse:
    def __init__(self, name) -> None:
        self.__spark = SparkSession.builder.appName(name).master("local[*]").getOrCreate()

        self.__df_product = None
        self.__df_categories = None
        self.__df_product_category = None

    @property
    def product(self):
        return self.__df_product

    @product.setter
    def product(self, data):
        self.__df_product = self.__spark.createDataFrame(data, schema=["id", "name"])
        self.__df_product.createOrReplaceTempView("products")

    @property
    def categories(self):
        return self.__df_categories

    @categories.setter
    def categories(self, data):
        self.__df_categories = self.__spark.createDataFrame(data, schema=["id", "name"])
```

```

self.__df_categories.createOrReplaceTempView("categories")

@property
def relation_prod2cat(self):
    return self.__df_product_category

@relation_prod2cat.setter
def relation_prod2cat(self, data):
    self.__df_product_category = self.__spark.createDataFrame(data, schema=["product_id", "category_id"])
    self.__df_product_category.createOrReplaceTempView("product_category")

def productInfo(self):
    return self.__spark.sql("""
        select p.name as `товар` , c.name as `категория`
        from products p
        left join product_category pc on p.id = pc.product_id
        left join categories c on c.id = pc.category_id
    """)

```

```

wh = Warehouse("warehouse1")

wh.product = products_data
wh.categories = categories_data
wh.relation_prod2cat = product_category_data

products_info = wh.productInfo()

products_info.show()

```

```

+-----+-----+
| товар|категория|
+-----+-----+
| Кофе|      NULL|
| Хлеб|      Еда|
| Молоко|    Еда|
| Яблоко|  Фрукты|
| Сок|  Напитки|
+-----+-----+

```

Тесты

```

product_name = [el[1] for el in products_data]
products_result = [row['товар'] for row in products_info.collect()]

product_name, products_result

```

```
(['Хлеб', 'Молоко', 'Яблоко', 'Сок', 'Кофе'],  
 ['Кофе', 'Хлеб', 'Молоко', 'Яблоко', 'Сок'])
```

```
for p in product_name:  
    assert p in products_result
```

```
"Проверка пройдена"
```

```
'Проверка пройдена'
```